

QUESTION 1

Preparation List

	Apparatus / Reagents / Chemicals	Quantity per candidate
1	In a capped container, labelled E , 10% urease solution	at least 15 cm ³
2	In a capped container, labelled U , 10% urea solution	at least 15 cm ³
3	In a beaker, labelled W , distilled water	At least 100 cm ³
4	Red litmus paper	One length approximately 20 cm or 4 X 5 cm strips from a book
5	Test-tube rack, suitable for holding 6 test-tubes	1
6	Test-tubes	6
7	Small beakers to hold up to 50 cm ³	6
9	10 cm ³ syringes	2
10	5 cm ³ syringes	2
11	White tile or paper or card	1
15	Ruler (mm)	1
16	Container, labelled ' Waste '	1

Preparation of Solutions

E, 10% urease solution at room temperature. This is prepared by dissolving 10 g of urease active meal or three crushed tablets of urease (according to manufacturer's instructions) in a beaker with 50 cm³ of distilled water. Make up to 100 cm³ with distilled water. Mix well. The solution may remain cloudy.

U, 10% urea solution at room temperature. This is prepared by adding 10 g of urea to 80 cm³ of distilled water in a beaker. Make up to 100 cm³ with distilled water. Mix well.

Testing the activity of the urease

Before the examination, put a small piece of red litmus paper into a dry test-tube. Add 2 cm³ of **E** then 2 cm³ of **U** and start timing. Time how long it takes for the paper to start turning blue.

If this is longer than 5 minutes, increase the concentration of urea to 15%.

It is not necessary to inform the students that the concentration is different from that given in the question paper.

QUESTION 2

Per Student

Solution for Each Candidate	Volume / cm ³
0.1 mol dm ⁻³ potassium chloride solution, adjusted to pH 7.0 . Solution should be sufficient to cover one piece of <i>Tradescantia</i> (spiderwort) leaf, in a beaker or container labelled X .	25
0.1 mol dm ⁻³ sodium chloride solution, adjusted to pH 7.0 . Solution should be sufficient to cover one piece of <i>Tradescantia</i> leaf, in a beaker or container labelled as Y	25
0.1 mol dm ⁻³ potassium chloride solution, adjusted to pH 4.5 . Solution should be sufficient to cover one piece of <i>Tradescantia</i> leaf, in a beaker or container labelled as Z	25

Apparatus for each candidate should be clean. Syringe needles are not required and must not be given to candidates.

Apparatus for each candidate	Quantity
Pipette, plastic or glass with teat 1	1
Paper Towel	8
Forceps (blunt)	1
Glass rod	1
Scissors	1
Glass Marker Pen	1
Scalpel or sharp blade	1
Stopwatch	1
Safety Glasses / Goggles	1
Microscope	1
Microscope slides with cover slips	5
Access to a sink and tap water	

Leaf Tissues:

Leaves from *Tradescantia* spp. (common name: Wandering Jew; Spiderwort).



Prepare the leaves by immersing in their different solutions and exposed to light for at least an hour before dispensing out to students. There should be sufficient leaves to provide at least 3 per candidate.

